VIT
Vellore Institute of Technology

J Component (EPJ) - Assessment Rubrics

Course Name: Natural Language Processing **Course Code**: SWE1017

Semester: FALL Semester 2026 -2027 Team size: 2 or 3

Sample Real-World NLP domains and research Problems

Domain	Sample Research Problems	
Healthcare	Clinical Report Summarization using LLMs	Medical Question Answering using RAG
Education	Automatic Short-Answer Evaluation	AI Tutor with Personalized Learning Recommendations
Social Media	Fake News Detection	Hate Speech Detection in Code-Mixed Languages
Cybersecurity	Phishing Email Detection	Cyber Threat Intelligence Extraction from Security Reports
Finance	Financial News Sentiment Analysis	Intelligent Loan Document Analysis
Legal	Legal Judgment Summarization	Contract Clause Extraction and Risk Analysis
Agriculture	Crop Disease Diagnosis from Farmer Queries	Agricultural Advisory Chatbot
Smart Cities	Public Grievance Classification	Emergency Incident Detection from Social Media
E-Commerce	Fake Review Detection	Product Review Summarization
Human Resources	Resume Screening and Job Matching	Skill Gap Analysis from Resumes and Job Descriptions
Research & Academia	Research Paper Summarization	Citation Recommendation System
Customer Service	Domain-Specific Customer Support Chatbot	Automatic Ticket Classification and Routing
Speech & Multimodal AI	Speech Emotion Recognition	Lip Reading with Language Models
Multilingual NLP	Code-Mixed Sentiment Analysis	Low-Resource Machine Translation for Indian Languages
Document Intelligence	Intelligent Document Information Extraction	PDF Question Answering using RAG
Government & Public Services	Citizen Complaint Analysis	Policy Document Summarization
Manufacturing / Industry 5.0	Predictive Maintenance Report Analysis	Machine Log Anomaly Detection using NLP
Travel & Tourism	Intelligent Travel Assistant	Tourist Review Sentiment Analysis

Domain	Sample Research Problems	
Media & Entertainment	Movie Review Analysis	Automatic News Summarization
Large Language Models (LLMs)	Hallucination Detection in LLMs	Explainable Retrieval-Augmented Generation (RAG)

Rubrics for the J –Component:

Review Dates	Review	Content to be addressed
6th & 7th Aug 2026	REVIEW–I (20 Marks)	Title of the project, Objectives (max 2), Experimental design(architecture diagram)
9th & 10th Sep 2026	REVIEW–II (30 Marks):	Title of the project, objectives (as in review I), Experiments carried till date with protocols, Procedures and results obtained (60% of implementations)
8th & 9th Oct 2026	REVIEW – III (50 Marks)	Title of the project, Objectives (as in previous reviews), Experiments carried out and the results obtained with appropriate discussion on the results obtained (Analysis) and report (as Research paper)

TITLE OF PROJECT REPORT

A PROJECT REVIEW I REPORT

Submitted by

Name of the Candidate(S) - Register No.(s)

<Course _code – Course_name> (EPJ)

Project Guide
<NAME >
<Designation>
<School>

NAME OF THE DEGREE/BRANCH
IN
NAME OF THE SCHOOL



FALL Semester 2026-27

Review I Report Submission

Submit the document with following content:

1. Abstract
2. Problem statement and Objectives
3. Review of 5 papers per student (IEEE/Springer/Elsevier or other reputed journals – not before 2023)

   (i) Title, Author, Date of publication and Publisher Information

   (ii) Problem and Objectives

   (iii) Methodology Adopted

   (iv) Performance Limitations

Literature survey (each student should refer minimum 5 papers) - Existing Systems, Gap identified, Proposed Method, Datasets Description and sample data.

Literature survey

S.No.	Title of the Paper, Publisher And year	Problem addressed	Algorithms used	Performance metrics used	Performance attained	Limitations(Gap identified)	Scope for future work

4. Summary of Literature and Limitations and Your proposed alternative methodology and design

5. References (As per IEEE format)